



networks selective

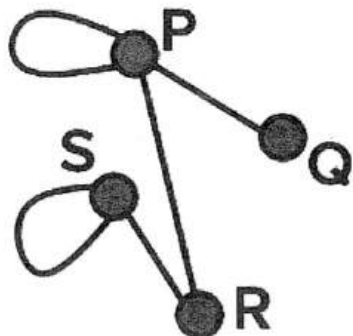
Standard

NAME: _____

Q 1

Mary Mackillop

What is the degree of the vertex marked P?



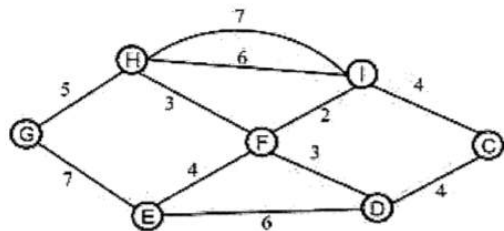
- (A) 1
(B) 2
(C) 3
(D) 4

D

Q 2

Mary Mackillop

The diagram below shows the seven suburbs of a city and the network of public transport links between them.
The amounts represent the cost (in dollars) of travel using that link.



What is the minimum cost of travelling from G to C?

- (A) \$14 (C) \$16
(B) \$15 (D) \$17

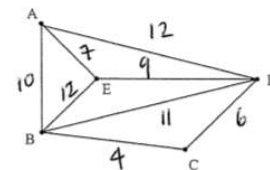
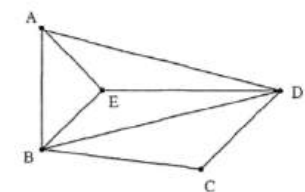
A

Q 3

Mary Mackillop

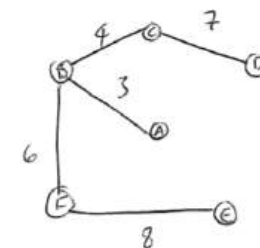
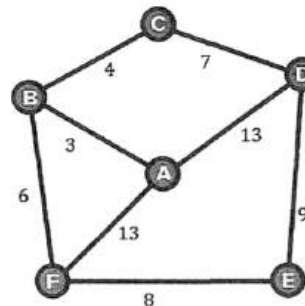
Complete a weighted network diagram from the information in the table.

	A	B	C	D	E
A	-	10	-	12	7
B	10	-	4	11	12
C	-	4	-	6	-
D	12	11	6	-	9
E	7	12	-	9	-



Q 4

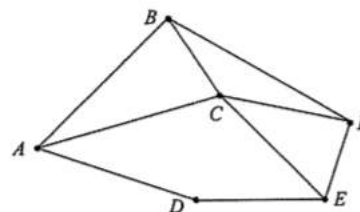
Mary Mackillop



Draw the minimum spanning tree for the network in the space below.

Q 5

ACE



Which of the following represents a circuit in the above network diagram?

- (A) B-A-D-E-C-B (B) D-A-C-F-D (C) E-F-C-A-B-E (D) C-F-E-D-C

A





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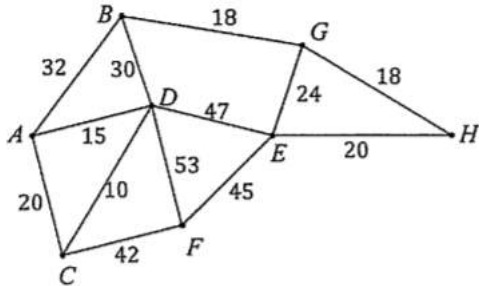
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Q 6

Mary Mackillop

Lucy owns a holiday park and has been quoted \$45 per metre to provide connecting paths to each of the cabins, as shown below. All measurements are in metres.



What is the least it will cost Lucy to provide each cabin with at least one path that connects all the cabins?

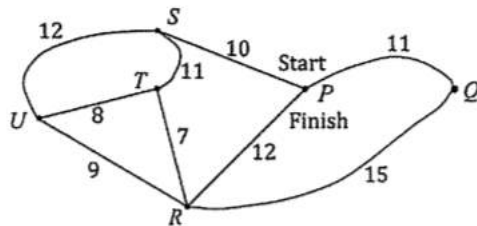
- (A) \$6435 (B) \$6885 (C) \$7470 (D) \$8820

B

Q 7

ACE

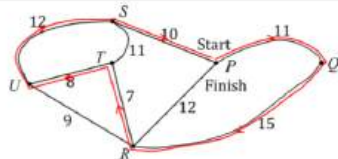
The network diagram below shows a running course. Competitors must pass through each of the checkpoints, P, Q, R, S, T and U. The average time (in minutes) taken to run between the checkpoints is shown on the edges of the diagram.



Competitors must start and finish at checkpoint P. However they can pass through the other checkpoints in any order they wish.

Which route would have the shortest average completion time?

Shortest path
P-Q-R-T-U-S-P
Length
= 11 + 15 + 7 + 8
+ 12 + 10
= 63 min

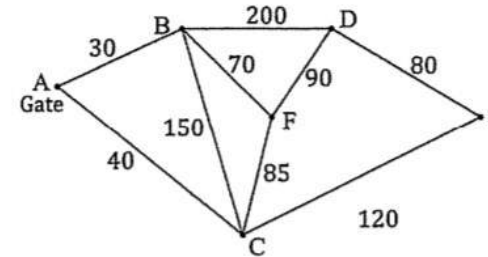


∴ Shortest average time is 63 minutes

Q 8

Mary Mackillop

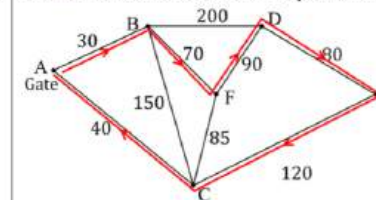
The network diagram below shows a park with 5 gardens that are joined by paths to the gate. The numbers show the distance along the paths in metres.



- (a) What is the degree of each vertex?
(b) What is the minimum distance a person would have to walk to visit every garden, starting and ending at the gate.
(c) Each garden needs to be connected by a water pipe located along a path. Draw a minimum spanning tree to indicated the location of the water pipes.

37(a) $\deg(A) = 2, \deg(B) = 4, \deg(C) = 4, \deg(D) = 3, \deg(E) = 2$ and $\deg(F) = 3$

37(b) Minimum distance is the shortest path to and from the gate.

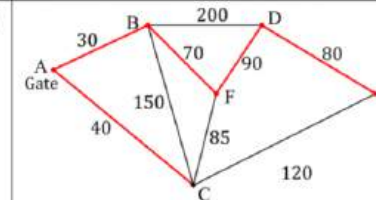


$$\text{Distance} = 30 + 70 + 90 + 80 + 120 + 40 \\ = 430 \text{ m}$$

∴ Minimum distance to visit every garden is 430 m

Note: other answers are possible.

37(c)





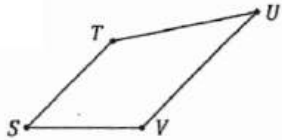
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Q 9

ACE



Which of the following walks is a path in the above network diagram?

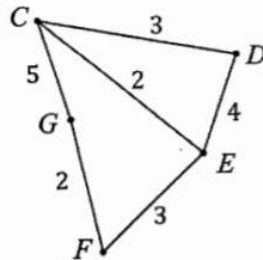
- (A) S-T-S-V
- (B) S-T-U-V
- (C) S-T-V-S
- (D) S-T-U-V-S

B

Q 9

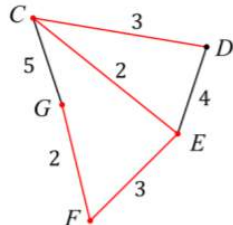
ACE

The network diagram below shows the possible paths (in km) for laying gas pipes between five locations.



What is the minimum length of pipes required to provide gas to all locations?

Find the minimum spanning tree.

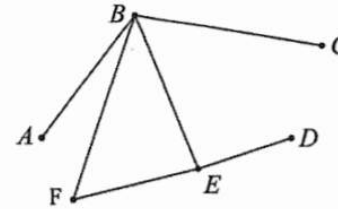


$$\begin{aligned} \text{Length} &= 2 + 3 + 2 + 3 \\ &= 10 \text{ km} \end{aligned}$$

$\therefore$ Minimum length of pipes is 10 km.

Q 10

ACE



(a) Complete the table of vertex degrees for the network diagram.

Vertex	A	B	C	D	E	F
Degree						

(b) Is there a path in the network that visits every edge exactly once? Give a reason for your answer.

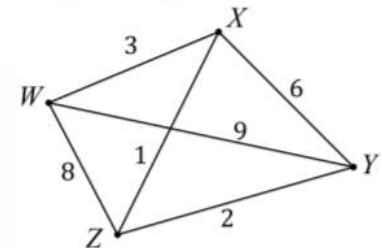
21(a)	<table><tr><td>Vertex</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td>Degree</td><td>1</td><td>4</td><td>1</td><td>1</td><td>3</td><td>2</td></tr></table>	Vertex	A	B	C	D	E	F	Degree	1	4	1	1	3	2
Vertex	A	B	C	D	E	F									
Degree	1	4	1	1	3	2									
21(b)	No. Path (Eulerian trail) only exists if the graph has exactly two vertices with an odd degree. There are 4 vertices with odd degree.														

Q 11

ACE

	W	X	Y	Z
W	-	3	9	8
X	3	-	6	1
Y	9	6	-	2
Z	8	1	2	-

Represent the table shown above as a weighted network.





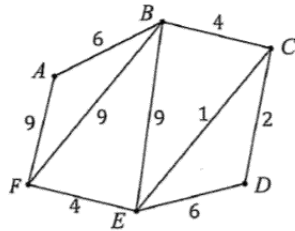
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Q 12

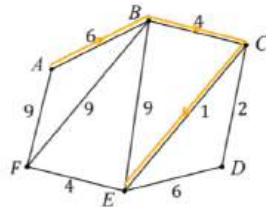
ACE



- (a) Find the length of the shortest path from A to E.
(b) Find a walk that visits every edge of the network only once, starting at C.

- 44(a) Identifying the alternative routes from A to E.
Shortest path is A-B-C-E

$$\text{Length} = 6 + 4 + 1 = 11$$



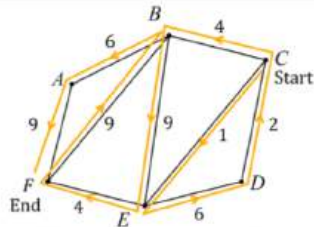
- 44(b) There are two vertices with an odd degree.

$\deg(C) = 3$ and $\deg(F) = 3$.

Walk

C-E-D-C-B-E-F-B-A-F

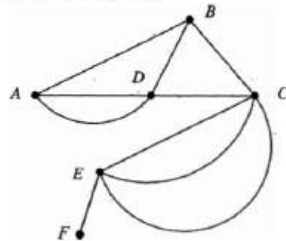
Note: other answers are possible.



Q 13

ACE

The diagram below shows a network.



- (i) Complete the table of vertex degrees using the network diagram.

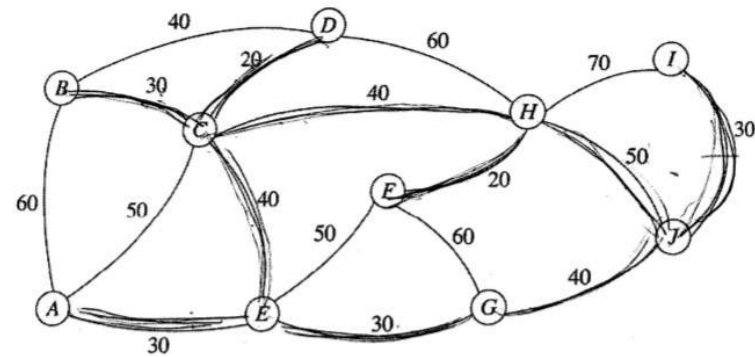
Vertex	A	B	C	D	E	F
Degrees						

- (ii) Explain why a Eulerian trail cannot be formed.

Q 14

Arthur Phillip

Once upon a time in the small village of Everton the pathways between its 10 houses always became very muddy in wet weather. The villagers decided to pave some of the paths with cobblestones so that all householders could travel to everybody's house on cobblestones without walking in mud.



The distance between the houses is shown on the diagram, in metres, and 30 cobblestones are used for every metre of road. If every cobblestone has an average mass of 7 kg, find the minimum mass of cobblestones required to complete the project.





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e) Distance

$$= 30 + 30 + 20 + 40 + 40 + 30 + 20 + 40 + 30$$

$$= 280$$

mass of stones

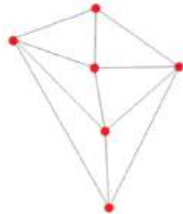
$$= 280 \times 30 \times 7$$

$$= 58800 \text{ g}$$

Q 15

Ascham

Which of the following is the spanning tree of the network below?

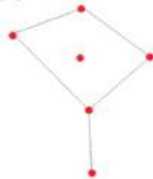


(C)

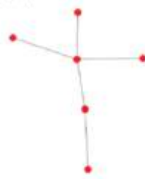
(A)



(B)



(C)



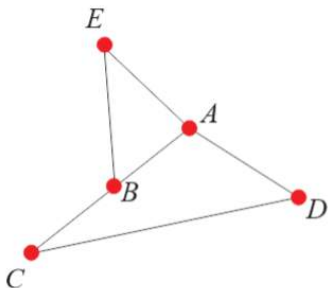
(D)



Q 16

Ascham

For the network below, *BADCBEA* represents:



(A) A path

(B) An Eulerian Circuit

(C) An Eulerian Trail

(D) A cycle

(C) An Eulerian Trail

Q 17

Ascham

Allison uses a dishwasher with a rating of 800 W for 2.5 hours each day. Find the daily cost of electricity for using the dishwasher if electricity is charged at a rate of 21.40 cents per kWh.

$$800 \text{ W} = 0.8 \text{ kW}$$

$$\text{Energy used} = 0.8 \times 2.5$$

$$= 2 \text{ kWh}$$

$$\text{Daily cost} = 2 \times 0.214$$

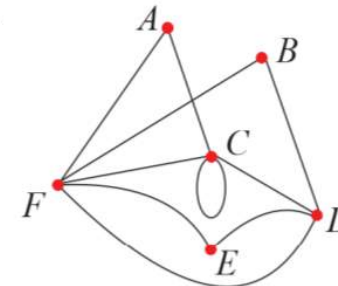
$$= 0.428$$

$$= \$0.43 \text{ (nearest c)}$$

Q 18

Ascham

Consider the network below.



- State the number of edges in the network.
- Find the sum of the degrees of the vertices in the network?
- Write one path that includes all of the vertices.
- Does this network have an Eulerian trail? Explain why.
- Find one spanning tree for this network.





networks selective

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NAME: _____

i) State the number of edges in the network.

10

ii) Find the sum of the degrees of the vertices in the network?

$$\text{Sum of degrees} = 2 + 2 + 5 + 5 + 2 + 4 = 20$$

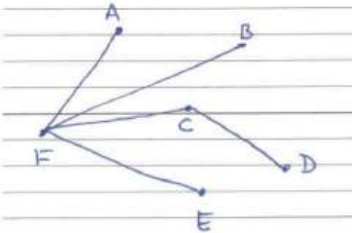
iii) Write one path that includes all of the vertices.

A path is a walk that doesn't visit any vertex more than once.
eg. ACDEFB

iv) Does this network have an Eulerian trail? Explain why.

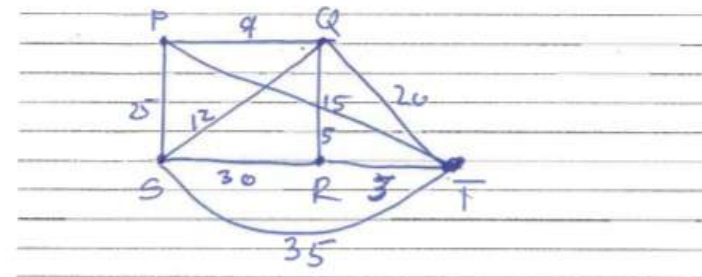
Yes, as this network has exactly two odd vertices F and C.

v) Find one spanning tree for this network.



i) Draw a weighted network that is represented by the given table.

[2]



ii) Find the minimum spanning tree.

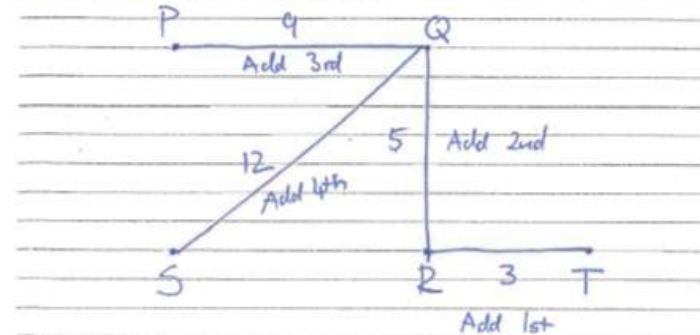
[2]

You can use Kruskal's or Prim's algorithm.

Kruskal's algorithm: List the edge weights.

RT	QR	PQ	QS	PT	QT	SP	RS	ST
3	5	9	12	15	20	25	30	35

Choose all of the vertices and add edges so they don't form a cycle.



Q 19

Ascham

The table below is used to represent a network.

		TO				
		P	Q	R	S	T
FROM	P	-	9	-	25	15
	Q	9	-	5	12	20
	R	-	5	-	30	3
	S	25	12	30	-	35
	T	15	20	3	35	-

i) Draw a weighted network that is represented by the given table.

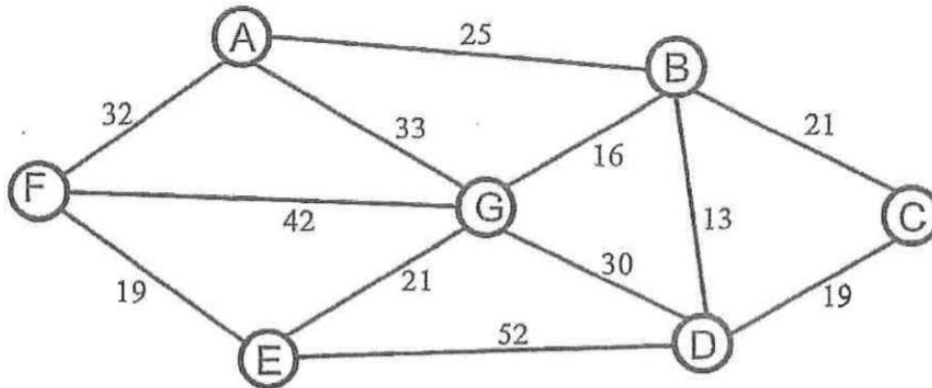
ii) Find the minimum spanning tree. Additional space provided on the next page.



Q 20

Barker

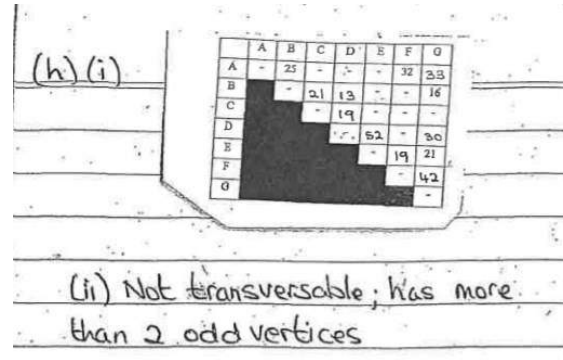
The network diagram below shows the distances between seven villages in a valley.



(i) Complete the unshaded spaces in the table to represent this network.

	A	B	C	D	E	F	G
A	-	25	-	-	-	32	
B		-			-	-	16
C			-		-	-	
D				-		-	
E					-		21
F						-	
G							-

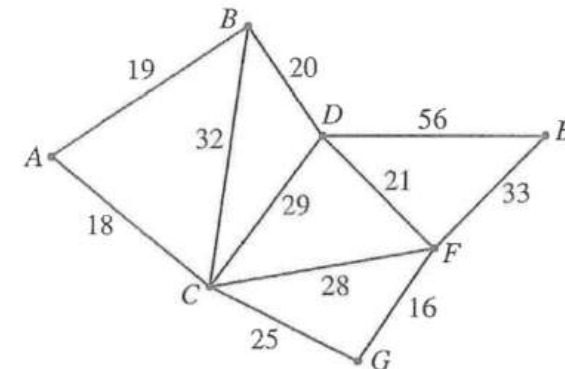
(ii) Is the network traversable? Justify your answer.



Q 21

Barker

The network below shows the road distances in kilometres between seven towns labelled A, B, C, D, E, F, G. To establish a cable network for communications, it is proposed to put the cable underground beside the existing roads.



- Show the minimum spanning tree on the diagram (by highlighting or emboldening) that ensures all the towns are connected to the network and minimises the amount of cable used.
- What is the minimum length of cable required here if back-up links are not considered necessary; that is, there are no loops in the cable network?
- A worker is at town A and needs to get to town E. Determine the shortest path.

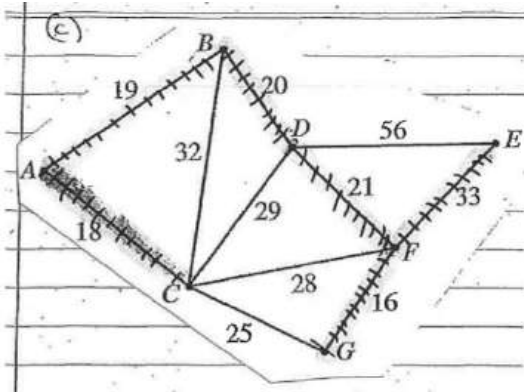


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(i) see above for minimum spanning tree.

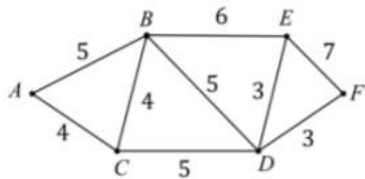
(ii) $16 + 18 + 19 + 20 + 21 + 33 = 127 \text{ km}$

(iii) shortest path is
 $A - C - F - E$ (79 km)

Q 22

Westfield Sports

What is the length of the shortest path from A to F in the Network Diagram below?

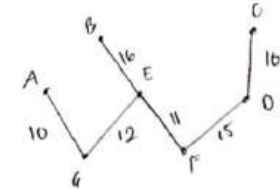
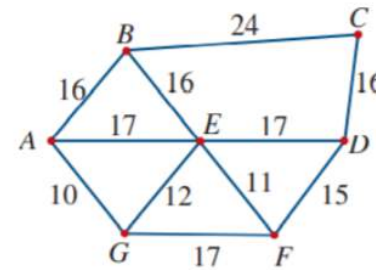


- | | |
|-------|-------|
| A. 12 | C. 14 |
| B. 13 | D. 15 |

Q 23

Westfield Sports

The diagram represents a network with weighted edges.



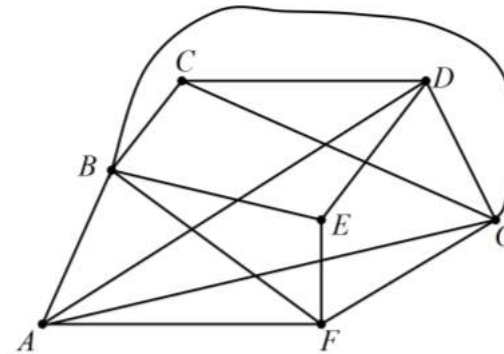
Minimum length of spanning tree = 80

Draw a minimum spanning tree for this network in the space below and determine its length.

Q 24

Total

Consider the network diagram below.



Which of the following is a cycle?

- A. ABEF
B. ABCDEFG
C. ABCDGAB
D. FEDGCBF

D

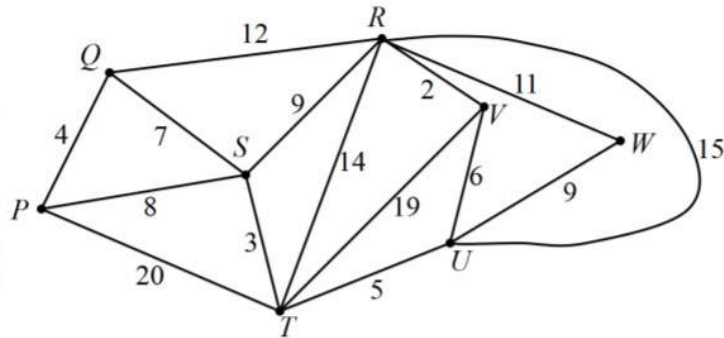




Q 25

Total

The graph of a university grounds is pictured below, along with the time taken to walk from building to building.



- How many nodes in the graph have an even degree?
- Calculate the fastest route to get from building P to building V .
- A university student wants to put up pamphlets at each building. Write down a path she could take, without walking along any particular road more than once.

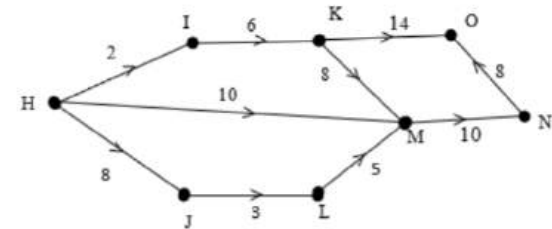
a **Sample answer:** 4 – S, R, U, W
b **Sample answer:** $PQRV$ – 18 minutes

c **Sample answer:** $PQSTRVUW$ or $TPSQRVUW$ or $VUWRTSQP$ or $VRWUTSQP$ or $QSPTRVUW$ or $VUWRSQPT$ (many possible solutions)

Q 26

Total

Below is a network flow diagram.



What is H an example of?

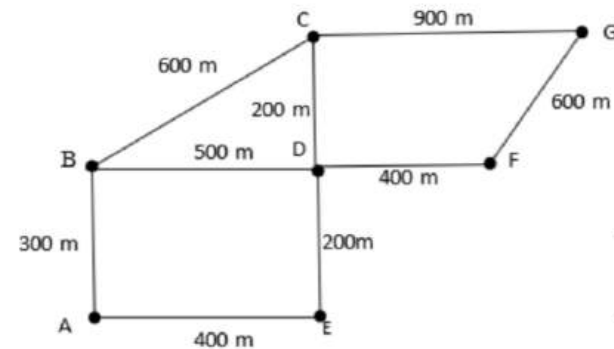
- Sink
- Path
- Edge
- Source

D

Q 27

Total

The diagram below shows the proposed road system for a new neighbourhood.



NOT TO SCALE

The council wishes to lay power cable that will connect all points in neighbourhood.

By drawing a minimum spanning tree, calculate the minimum length of cable required.

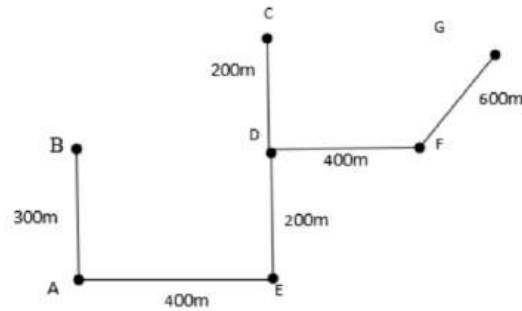


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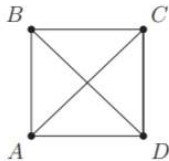


Total distance =
 $300 + 400 + 200 + 200 + 400 + 600 = 2100 \text{ m or } 2.1 \text{ km}$

Q 28

Sydney Grammar

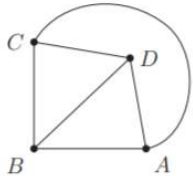
A network of four points A, B, C and D is drawn below.



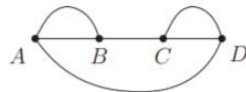
B

Which of the following network diagrams is NOT an equivalent graph of the network above?

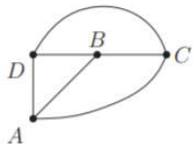
A.



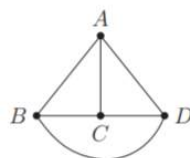
B.



C.



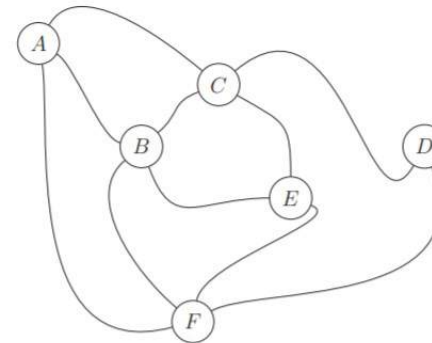
D.



Q 29

Sydney Grammar

A collection of bushwalking tracks is modelled by the network below. There are six checkpoints labelled A, B, C, D, E and F . Which of the following two edges should be removed so that a bushwalker could walk along every track exactly once?



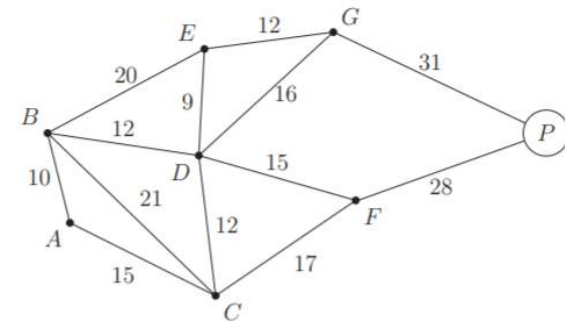
A

- A. AB and BE B. CB and BF C. CD and DF D. CE and BF

Q 30

Sydney Grammar

Cabins in a national park are to be connected to the power source at P . The weighted graph below shows the distances in metres between certain cabins and the power source.



- (a) Use Kruskal's algorithm to draw a minimum spanning tree for this network.
(b) What is the weight of this minimum spanning tree?



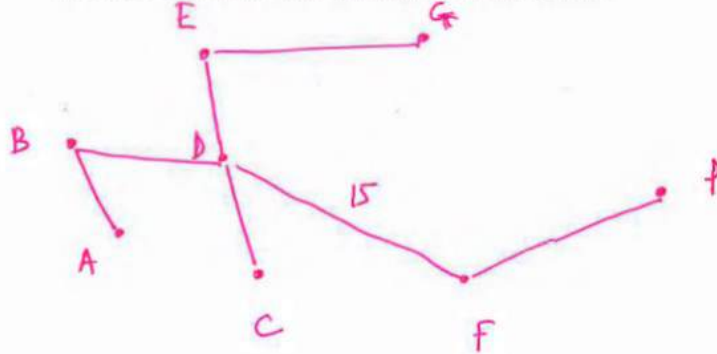
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(a) Use Kruskal's algorithm to draw a minimum spanning tree for this network.

$ED=9$ ✓
 $BA=10$ ✓
 $BD=12$ ✓
 $EG=12$ ✓
 $DC=12$ ✓
 $DF=15$ ✓
 $AC=15$ ✗
 $DG=16$ ✗
 $CF=17$ ✗
 $BE=20$ ✗
 $FP=28$ ✓
 $GP=31$ ✗



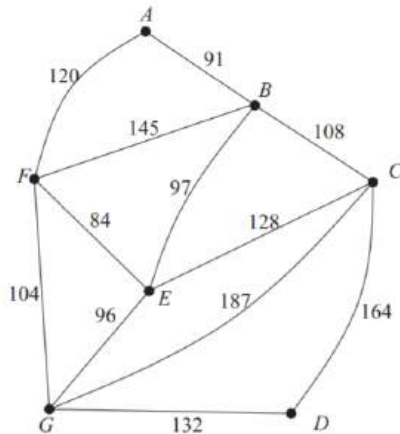
(b) What is the weight of this minimum spanning tree?

$$10 + 12 + 9 + 12 + 12 + 15 + 28 = 98$$

Q 31

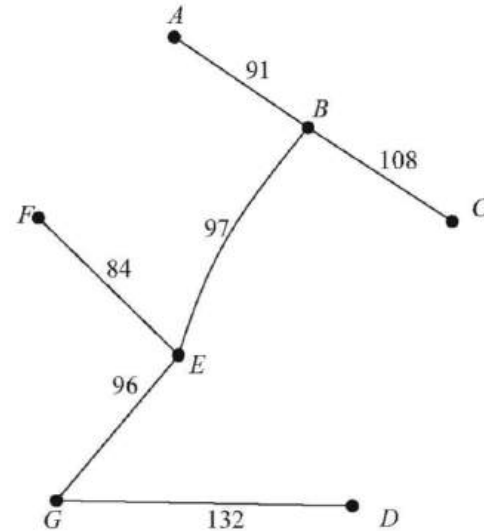
Sydney Grammar

The network diagram shows the electrical wires connecting 7 buildings on a school campus. The vertices A to G represent the buildings. The weights on the edges represent the length of wire in metres used to connect the buildings.



(a) Draw the minimum spanning tree in the space below.

(b) What is the minimum length of wire required to connect all the buildings with electricity?

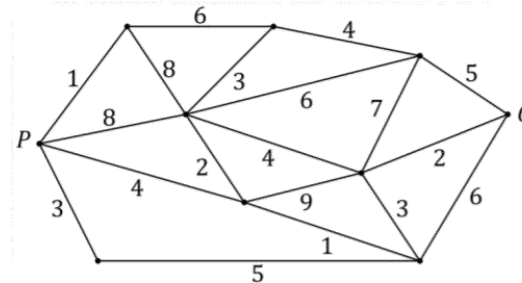


608 metres

Q 32

St George Girls

The following network shows the time, in hours, that it takes to travel along a series of roads that connect town P to town Q.



(C) 10

What is the shortest time, in hours, that it would take to travel from town P to town Q?

(A) 8

(C) 10

(B) 9

(D) 14





Q 33

St George Girls

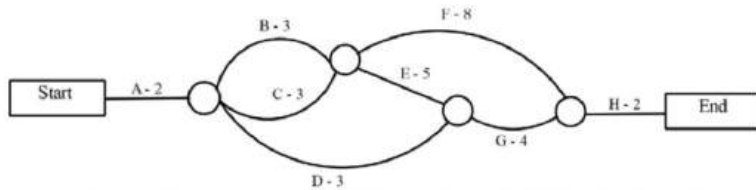
The activity chart below shows the immediate prerequisite(s) and duration for each activity in a project.

Activity	Immediate Prerequisites	Time (days)
A	-	2
B	A	3
C	A	3
D	B, C	3
E	A	5
F	B, C	8
G	D, E	4
H	F, G	2

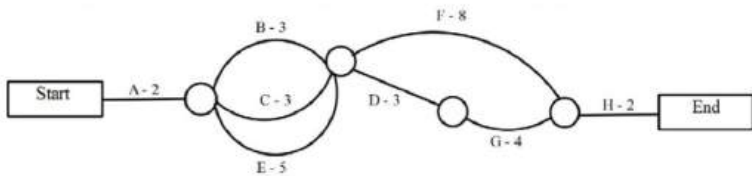
(C)

Which network could be drawn from the activity chart?

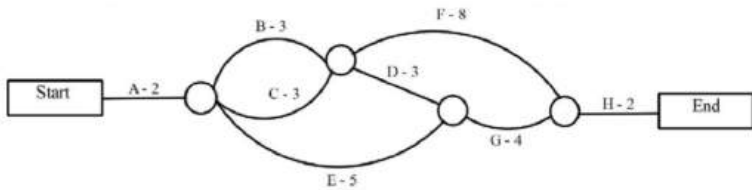
(A)



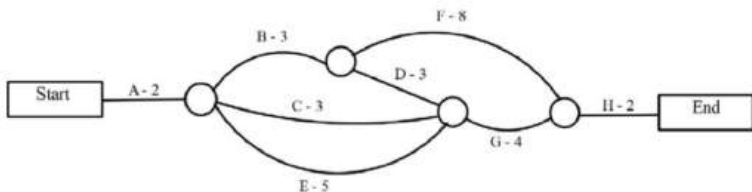
(B)



(C)



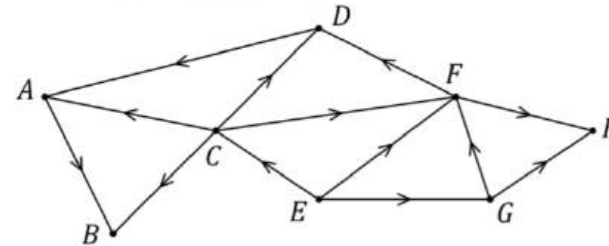
(D)



Q 34

St George Girls

A maze contains eight statues. The statues are labelled A to H on the following directed graph. Walkers within the maze are only allowed to move in the directions of the arrows.



(a)

E & C

(b)

EFD, EGFD

(a)

Find two statues that a walker could not reach from statue G.

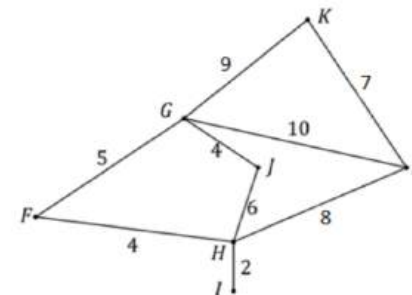
(b)

One way that statue D can be reached from statue E is along path ECD. List two other ways that statue D can be reached from statue E.

Q 35

St George Girls

The network diagram shows seven campsites, F, G, H, I, J, K and L, which are joined by tracks. The numbers by the paths show lengths (in km) of that section of track.



(a)

Find and highlight the minimum spanning tree of the network.

(b)

A telephone cable is to be laid along as few of the existing tracks as possible. What is the minimum length of cable necessary to complete this task?

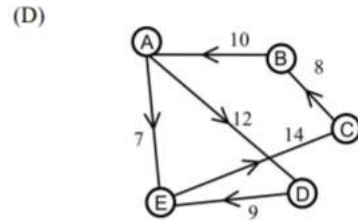
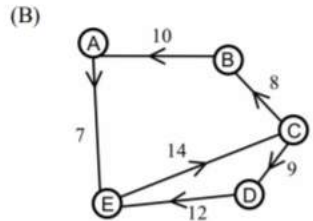
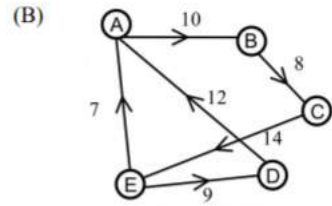
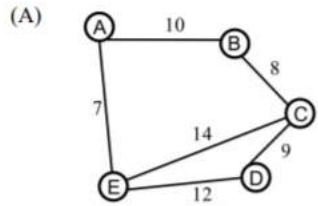


Q 36

Western

Which network would be described by the table below?

		To				
		A	B	C	D	E
From	A	-	-	-	12	7
	B	10	-	-	-	-
	C	-	8	-	-	-
	D	-	-	-	-	9
	E	-	-	14	-	-

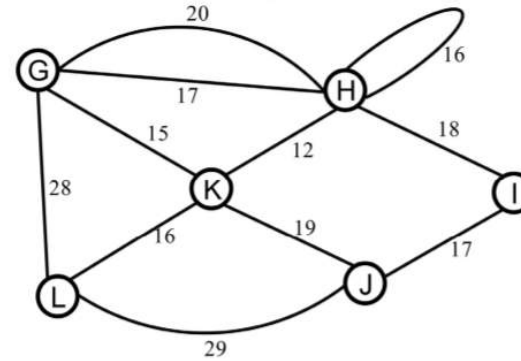


D

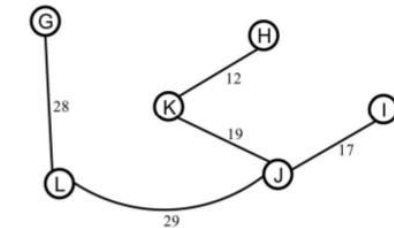
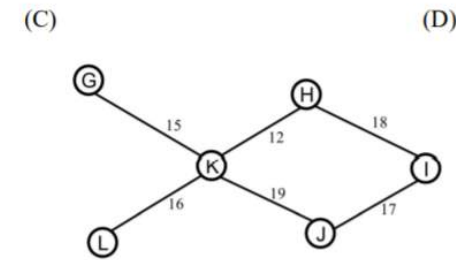
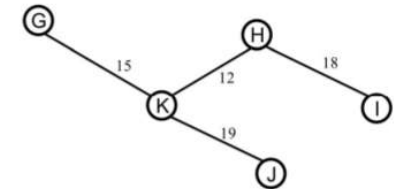
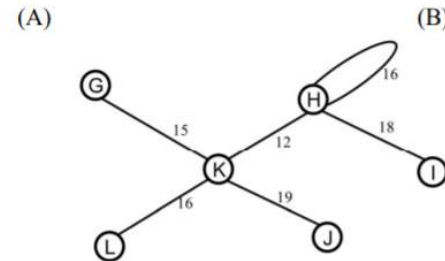
Q 37

Western

Which diagram shows a spanning tree for the network below?



D



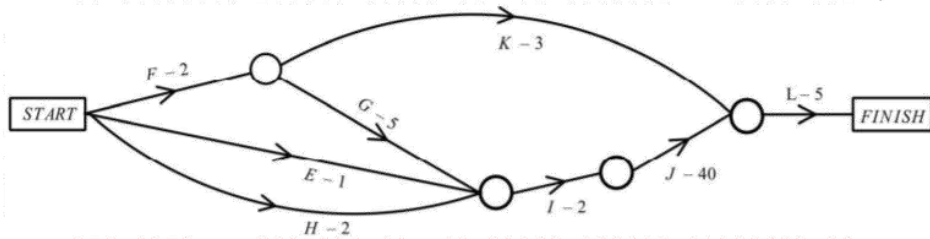
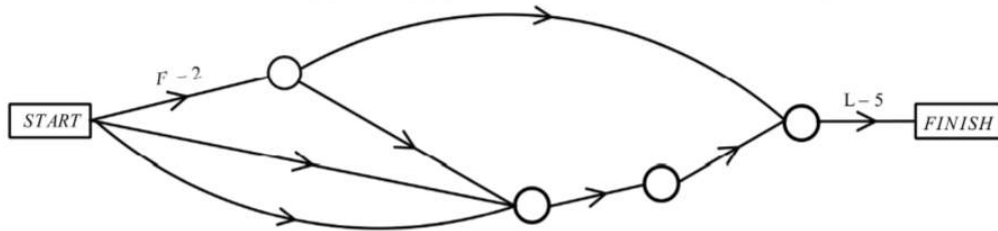
Q 38

Western

The table below outlines the steps in producing and harvesting a crop of grain.

Activity	Description	Time (days)	Prerequisites
E	Purchase Seed	1	-
F	Service Tractor and Plow	2	-
G	Plow Field	5	F
H	Apply Fertiliser	2	-
I	Plant Seed	2	E, G, H
J	Wait for Crop to grow	40	I
K	Service harvester	3	F
L	Harvest Crop	5	K, J

Complete the network diagram below, by writing the remaining labels on the edges.

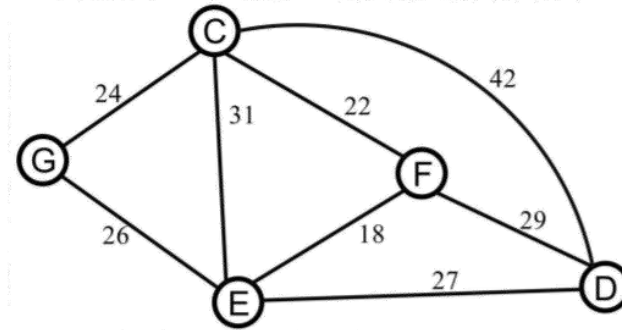


Q 39

Western

There are five towns that are connected by a series of roads.

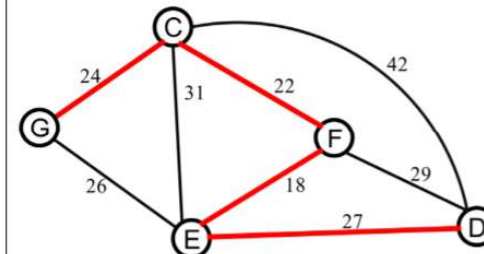
The distance between each town is shown on the diagram.



A funding program is introduced to re-seal some of the roads so that it is possible to travel between any two towns (not necessarily directly) along a newly sealed road.

- What name, in network terminology, is given to the path which meets this requirement?
- What is the least length of road that would need to be re-sealed?

- | | |
|-----|-------------------------|
| (a) | A minimum spanning tree |
| (b) | Using Prim's Algorithm |



Select C as the first vertex (arbitrary as any other could be chosen)

Highlight CF as lowest weight leaving C

Highlight FE as lowest weight leaving the tree.

Highlight CG as lowest weight leaving the tree.

Highlight ED as lowest weight leaving the tree not creating a cycle.

Every vertex is now included, so we have our minimum spanning tree

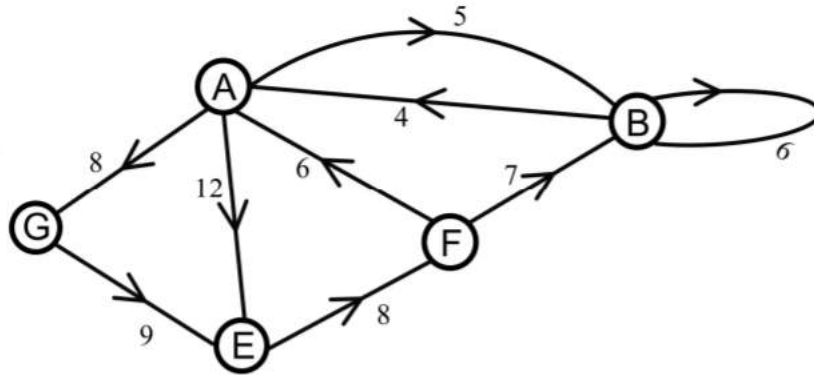
Length of road to re-seal = $24 + 22 + 18 + 27 = 91$ km



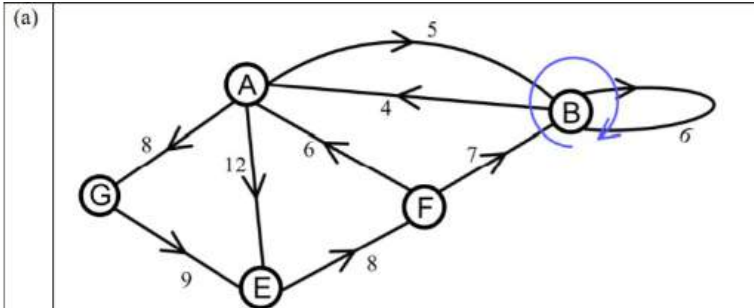
Q 40

Western

Alan drew the network shown below.



- (a) What is the degree of the vertex B?
- (b) Alan describes the network as follows:
"A disconnected, directed network which includes two loops."
Explain which parts of his description are incorrect.



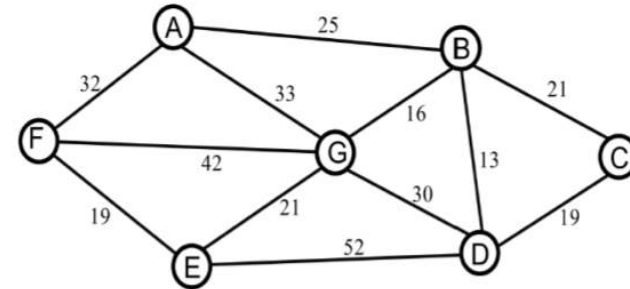
The circle around B passes through 5 edges, so the degree is 5

- (b) **Incorrect**
It is not disconnected as every vertex can be reached from any other vertex.
It has only one loop, not 2, which is at B with weight 6.
Correct
It is a directed network

Q 41

Blacktown

The network diagram below shows the distances between seven villages in a valley.



- (a) Complete the table to represent the network.

	A	B	C	D	E	F	G
A	-	25	-	-	-	32	33
B	25	-	21	13	-	-	16
C	-	21	-	19	-	-	-
D	-	13	19	-			
E	-	-	-	52			
F	32	-	-	-			
G	33	16	-	30			

- (b) Describe the shortest path between F and D, and calculate its distance.





networks selective



NAME: _____

Standard

(a)

	A	B	C	D	E	F	G
A	-	25	-	-	-	32	33
B	25	-	21	13	-	-	16
C	-	21	-	19	-	-	-
D	-	13	19	-	52	-	30
E	-	-	-	52	-	19	21
F	32	-	-	-	19	-	42
G	33	16	-	30	21	42	-

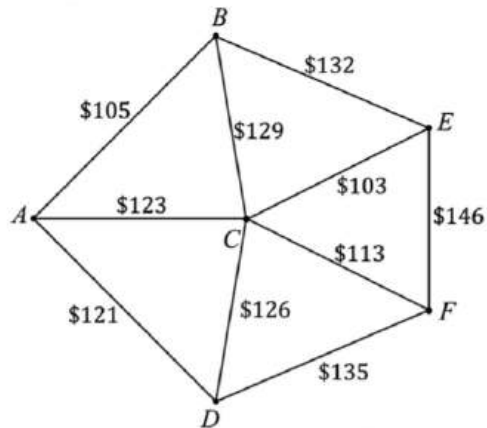
(b)

FE, GD ①
 $19 + 21 + 16 + 13 = 69$ ①

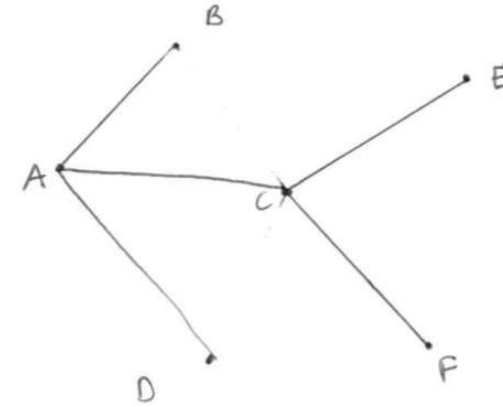
Q 42

Blacktown

The network diagram below shows the cost to lay pipes to certain parts of a garden.



- (a) Draw a minimum spanning tree that will ensure all parts of the garden are connected by pipes, but also minimises the amount of pipes required.
- (b) What is the minimum cost of pipes to connect all parts of the garden?

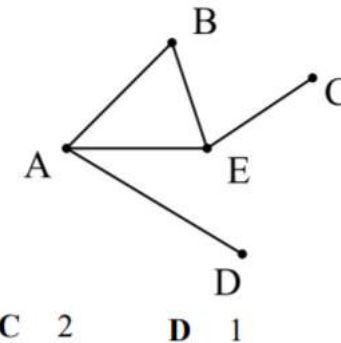


$105 + 121 + 123 + 103 + 113 = 565$

Q 43

Blacktown

What is the degree of vertex A in this network diagram?



A 4 B 3 C 2 D 1

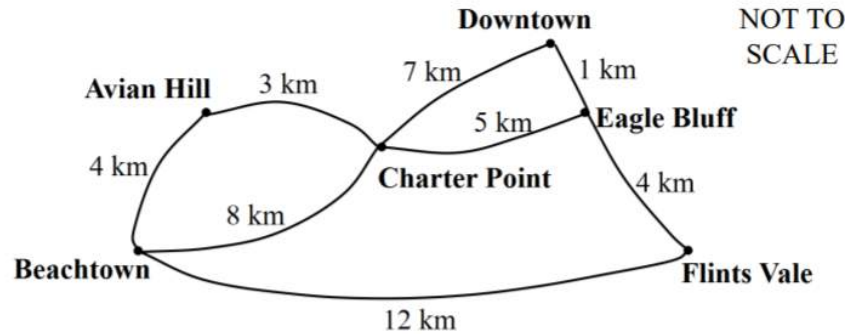




Q 44

Manly

This network diagram shows the streets connecting 5 suburbs.



What is the shortest possible route to get from Beachtown to Downtown?

- A 13 km C 15 km
B 14 km D 16 km

A

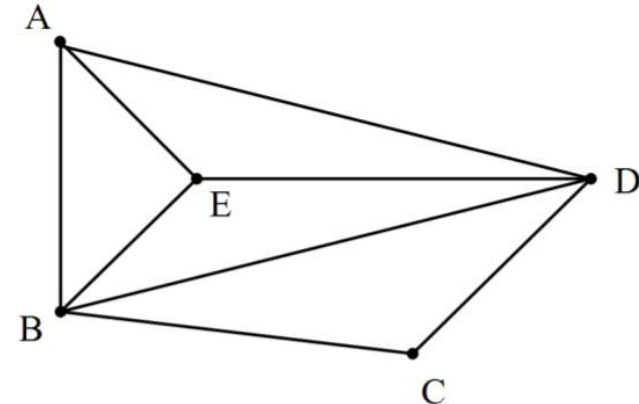
Q 45

Manly

A university IT department has done a study to analyse the cost of installing a new high-speed fibre optic network on the campus, taking into account the distances and current infrastructure. The cost, in thousands of dollars, to connect cables between the main buildings is provided in the table below. (Buildings that cannot feasibly be connected are indicated by a “-” symbol.)

	A	B	C	D	E
A		10	-	12	7
B	10		4	11	12
C	-	4		6	-
D	12	11	6		9
E	7	12	-	9	

- (a) Use the table above to complete the weighted network diagram below, representing the cost to connect the buildings in thousands of dollars.



- (b) Use your network diagram in part (a) to draw a minimum spanning tree for the campus in the space provided below.
- (c) What is the minimum cost (in dollars) to connect all the buildings on the campus with fibre optic cabling?



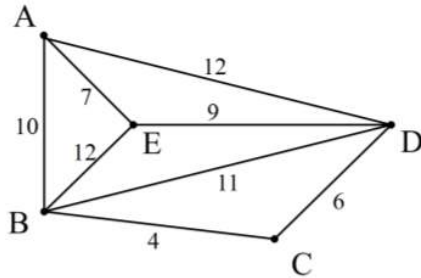
networks selective



NAME: _____

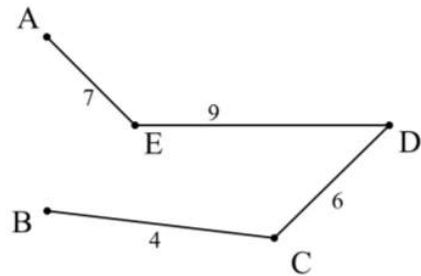
Standard

(a)



- ✓✓ All edges correctly labelled
- ✓ At least 4 edges correctly labelled

(b)



✓✓ Answer is isomorphic to the above network

✓ A spanning tree is drawn that is not the minimum spanning tree

(c)

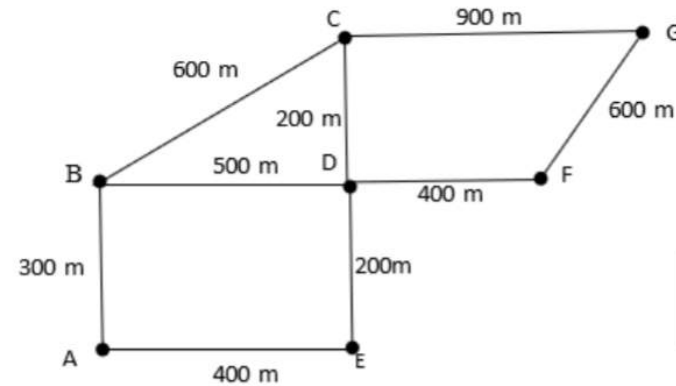
$$7 + 9 + 6 + 4 = 26 \checkmark$$

\$26,000 (Diagram represents thousands of dollars.)✓

Q 46

Manly

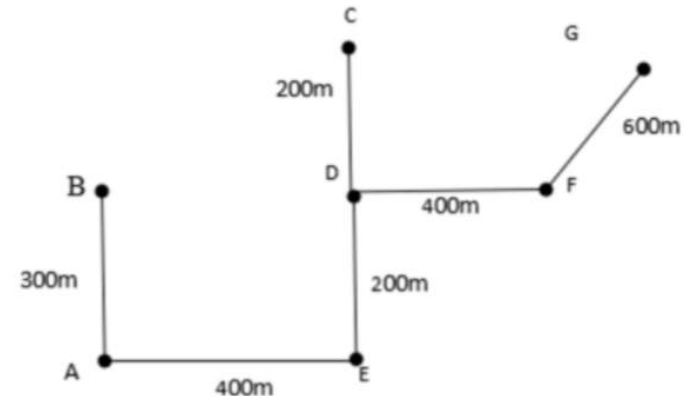
The diagram below shows the proposed road system for a new neighbourhood.



NOT TO
SCALE

The council wishes to lay power cable that will connect all points in the neighbourhood.

By drawing a minimum spanning tree, calculate the minimum length of cable required.



Total distance =

$$300 + 400 + 200 + 200 + 400 + 600 = 2100 \text{ m or } 2.1 \text{ km}$$



networks selective

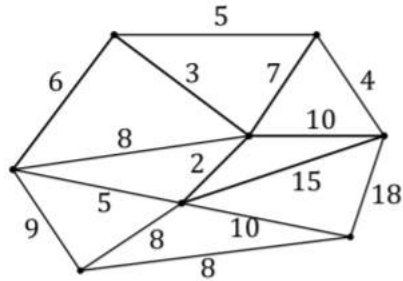


NAME: _____

Standard

Q 47

Manly

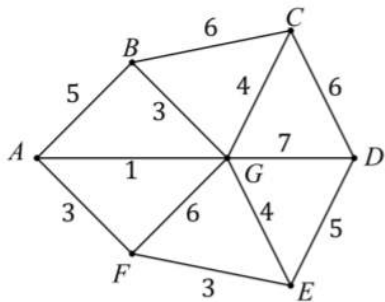


What is the length of the minimal spanning tree in the above network?

- (A) 30 (C) 39
(B) 35 (D) 45

Q 48

Cabramatta



The numbers in the above network represent time in minutes. Find the shortest time from vertex B to vertex F.

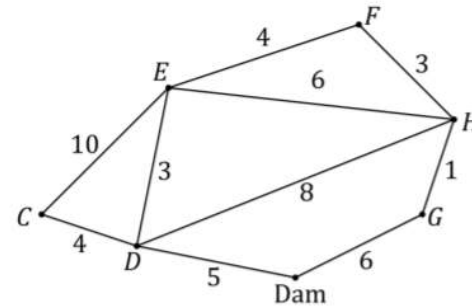
B - G - A - F

3 + 1 + 3 = 7 minutes

Q 49

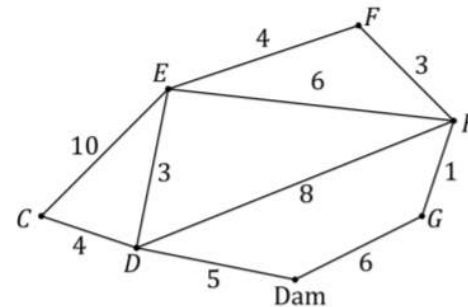
Cabramatta

The network diagram below shows pipes providing water from a dam to six farms. These pipes connect the farms to each other and to the dam. The lengths of the pipes (in km) are shown.



A plumber is hired to check the water pipes. He starts his examination at the dam, travels along each of the pipes and returns to the dam without travelling along any pipe section more than once.

- (a) Explain why such a path is possible for this network of pipes.
(b) Find a path the plumber could follow.
(c) Draw a minimum spanning tree on the network diagram below.



- (d) The system of pipes is due for replacement. What is the minimum length of pipe that can be used to service all of the farms?



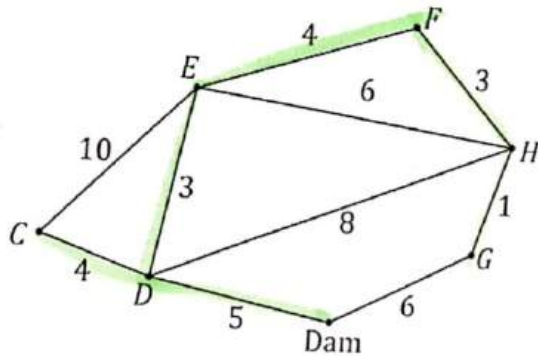
networks selective

Standard

NAME: _____

- (a) This network forms a cycle with 3 additional pipes on the inside which makes the network a Hamiltonian whereby all edges can be reached and vertices repeated.

- (b) Dam - D - C - E - D - H - E - F - H - G - Dam.



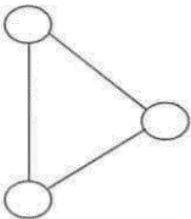
- (d) $4 + 3 + 4 + 3 + 1 + 5 = 20 \text{ km}$

Q 50

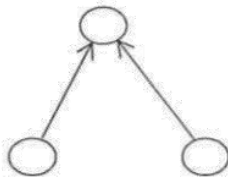
Shore

Which of the following are **not** connected networks?

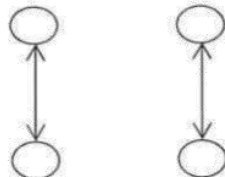
I



II



III



A. II

C. I and II

B. III

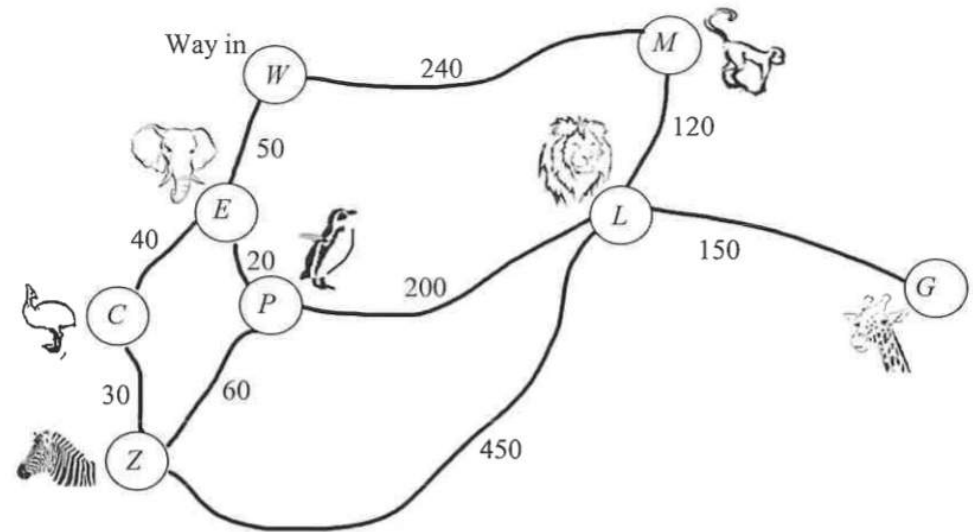
D. II and III

D.

Q 51

Shore

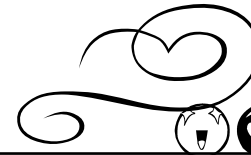
A map of Ziggy's Zoo is shown. Each animal enclosure and the way in are represented with a vertex. The weight on each edge is the distance from vertex to vertex, in metres.



- Draw a minimum spanning tree in the space below, with the correct weightings.
- The zoo needs to put in pipes to take sewage away from each animal enclosure. What is the shortest total length of pipe that will be needed to connect all of the animal enclosures (**not including the Way In**)?
- Find the shortest path from the Monkey enclosure (M) to the Zebra enclosure (Z).

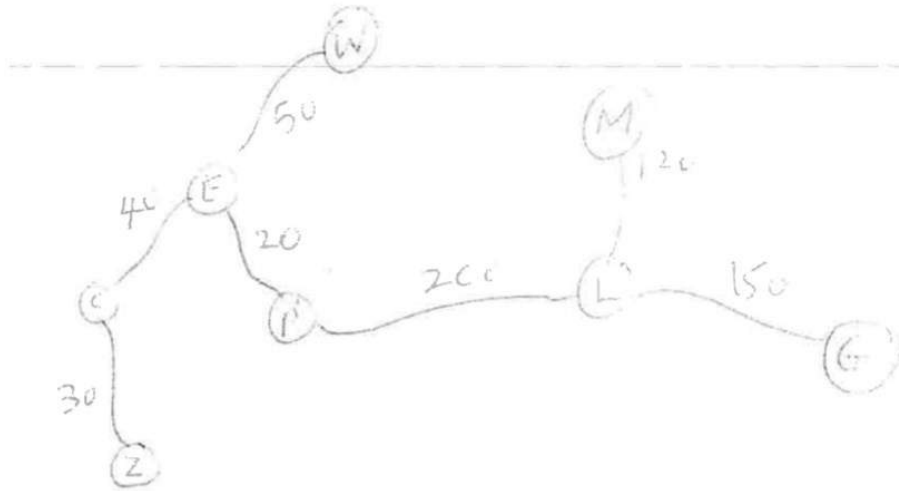


networks selective



NAME: _____

Standard



(ii)
$$\text{length} = 30 + 40 + 20 + 200 + 120 + 150$$

$$= 560 \text{ m}$$

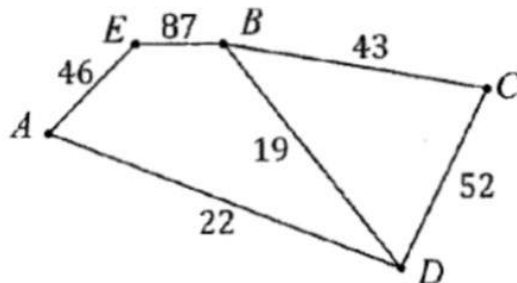
(iii) M W E C Z

Q 52

Cecil hills

What is the length of the shortest path from E to C?

- a) 117
- b) 120
- c) 138
- d) 158



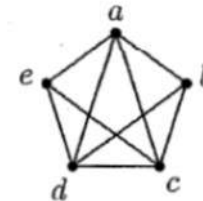
a) 117

b) 120

Q 53

Cecil hills

List the vertices of an Eulerian Trail using the network diagram.



Start and end at odd degree that is e NOT the same vertex!

All vertices must be covered.
ie. 10 Vertices + 9 edges.

Start a Stop e
OR start e stop a.

(Very few students got this right)

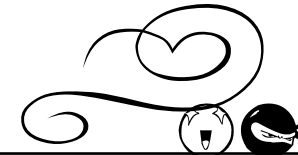
Q 54

Cecil hills

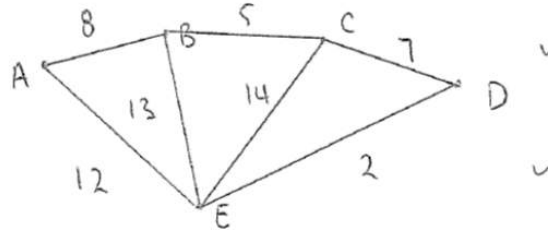
There are five towns labelled A, B, C, D and E. The table below shows which towns are linked by the roads and the length of each one in kilometres.

	A	B	C	D	E
A	-	8	-	-	12
B	8	-	5	-	13
C	-	5	-	7	14
D	-	-	7	-	2
E	12	13	14	2	-

- a) Represent the table shown above as a weighted network.
- b) What is the degree of vertex E?
- c) What is the shortest path from town A to town C?



a) Represent the table shown above as a weighted network.



b) What is the degree of vertex E?

4 ✓

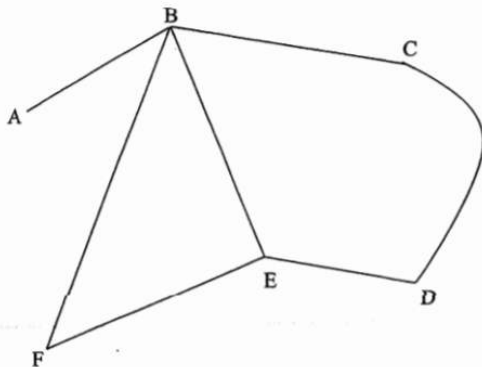
c) What is the shortest path from town A to town C?

A-B-C → 8+5 = 13 km ✓

Q 55

Cecil hills

Consider the following network diagram.



a) How many edges are there?

b) How many vertices have a degree of 2?

c) Is this a connected graph? Justify your answer.

a) How many edges are there?

7 ✓

• Done well

b) How many vertices have a degree of 2?

3 ✓

• Done well. Ensure you answer how many, not just list the vertices.

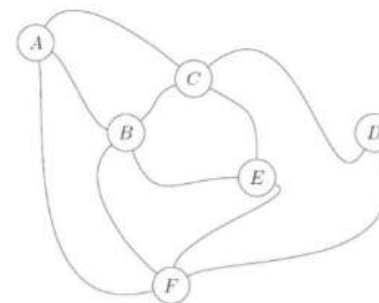
c) Is this a connected graph? Justify your answer.

Yes, as all vertices are connected to the other vertices through edges either directly or indirectly. ✓

Q 56

Caringbah

A collection of bushwalking tracks is modelled by the network below. There are 6 lookout points, labelled A, B, C, D, E and F. Which two edges should be removed so that a bushwalker can walk along every track exactly once and still visit every lookout?



(A) AB and BE

(B) CB and BF

(C) CD and DF

(D) CE and BF

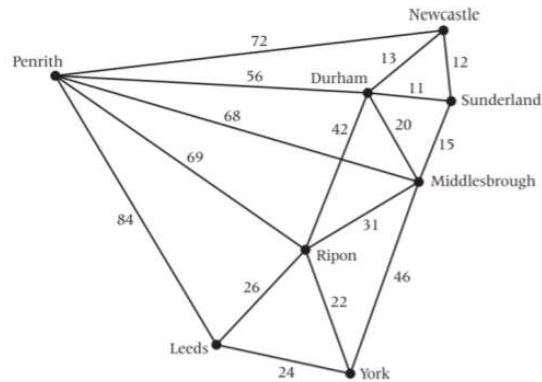
A



Q 57

Caringbah

This network shows the distance, in kilometres, between certain English towns



- What is the degree of the vertex at Middlesbrough?
- Iris lives in Penrith, she is going to pick up her friend in Newcastle and then drive to Leeds, find the shortest path, and its distance, to complete her trip

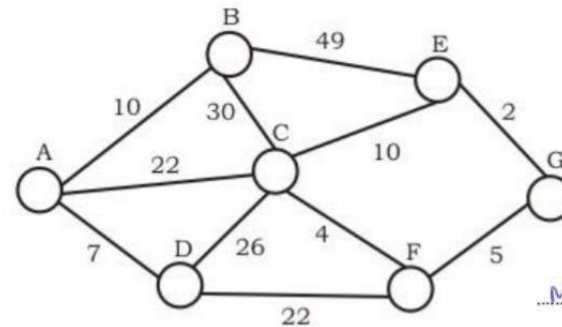
a) 5

b) P-D-N-D-R-L
150 km

Q 58

Caringbah

The following network shows the distance in metres between cabins (labelled A, B, C, D, E, F and G) in a holiday park. The park wants to run pipes between cabins using the existing roads shown in the network. If piping costs \$450/m, what is the minimum cost to run pipes to all of the cabins?

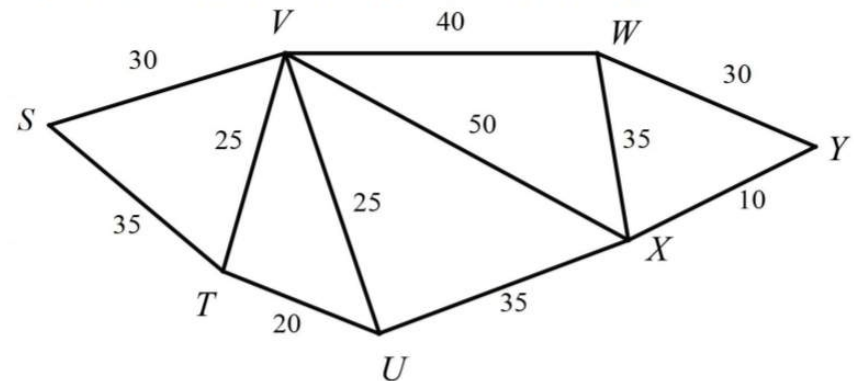


Minimum Spanning tree = 50m
50 x 450
= \$ 22500

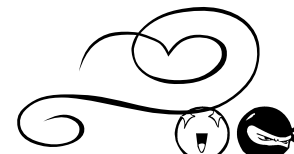
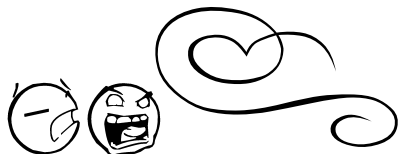
Q 59

Hurlstone

A new electricity supply network is to be constructed for seven buildings on a farm. The owners have considered possible routes for the cables connecting the buildings. The construction cost *in hundreds of dollars* is shown on each cable. The generator for the network can be located in any building.



- Draw a minimum spanning tree for the electricity network on the vertices
- Calculate the minimum cost of constructing the network of cables.

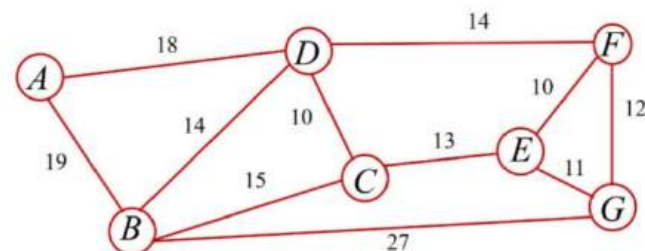


- a) Either of the following
-
- b) Sum of edges = $30 + 25 + 20 + 35 + 10 + 30 = 150$
Cost = \$15 000

Q 60

Hurlstone

The network diagram below represents a network of tracks joining buildings on a dairy farm.

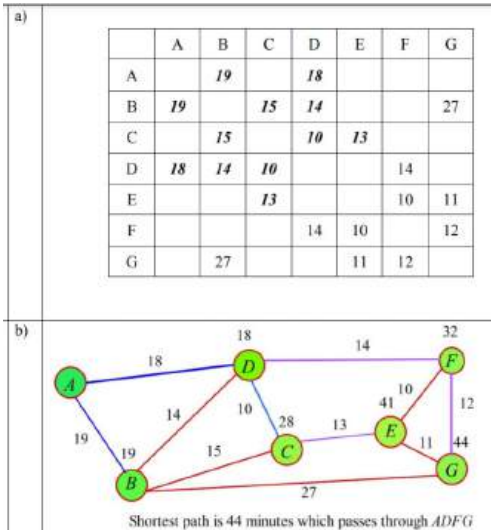
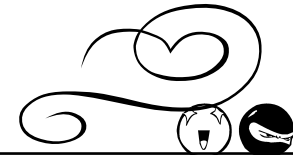


All tracks can be travelled in either direction.
The numbers indicate the travel time between buildings in minutes.

- (a) Complete the missing values in the table below to represent this network diagram.

	A	B	C	D	E	F	G
A							
B							27
C							
D						14	
E						10	11
F				14	10		12
G		27			11	12	

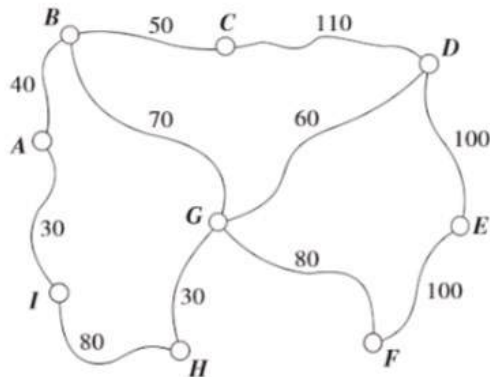
- (b) Determine the shortest travel time between buildings A and G and the tracks which would be followed to achieve this time.



Q 61

Fairvale

The map below shows the roads connecting Towns A to I. The value on the edges indicates the road distance in kilometres.



What is the shortest road distance from Town A to Town E?

- ☐ 300 km ☐ 270 km
☐ 260 km ☐ 290 km

Q 62

Fairvale

A project consists of seven tasks, A to G. The immediate predecessors and duration of each task are outlined below.

Activity	Immediate predecessor(s)	Duration (days)
A	—	6
B	—	7
C	A	3
D	C	2
E	A, D	1
F	B	6
G	E, F	4

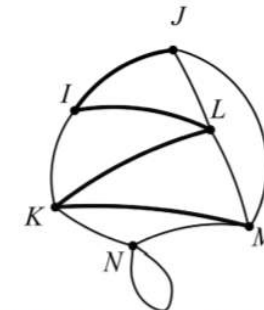
What is the minimum time required to complete the project?

- ☐ 18 days
☐ 16 days
☐ 15 days
☐ 17 days

Q 63

James Busby

An undirected network with vertices I, J, K, L, M and N is shown below.



Which term could **not** be used to describe the highlighted section J I L K M?

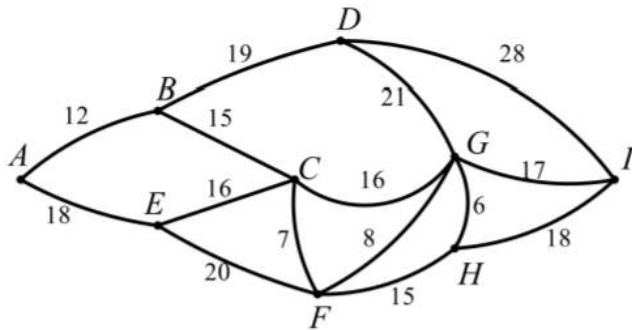
- (A) a cycle (C) a tree
 (B) a path (D) a walk



Q 64

James Busby

The network below shows the roads connecting nine villages labelled *A* to *I*.
The weights on the edges are the average travel times (in minutes) between adjacent villages.



What is the minimum travel time from *B* to *H*?

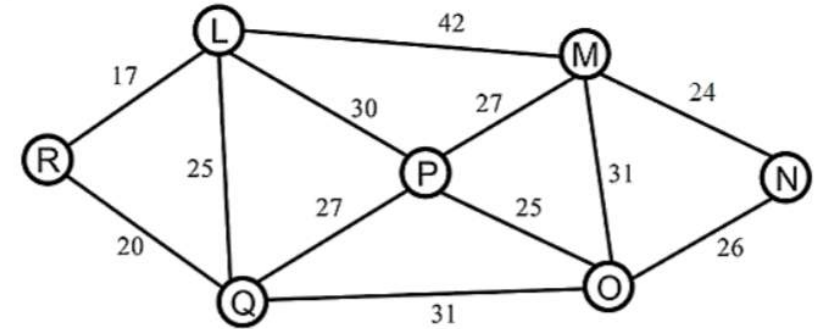
- (A) 31 minutes
- (B) 33 minutes
- (C) 36 minutes
- (D) 37 minutes

(C)

Q 65

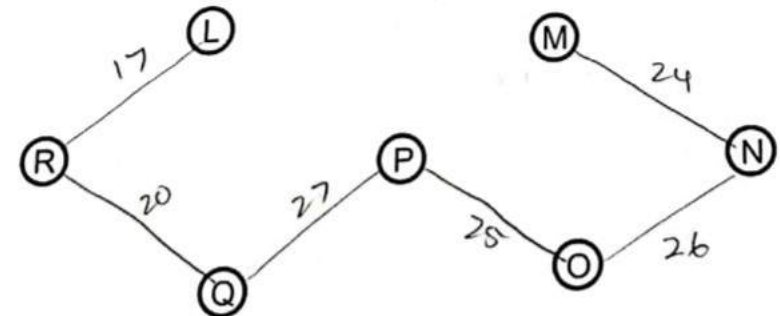
James Busby

The length of each pipe in kilometres is shown.



The owner of the network wants to reduce the number of pipes but still connect all the nodes.

- (a) Draw the minimum spanning tree that will achieve the owner's requirement on the set of nodes below.
- (b) What is the minimum length of pipe needed to connect the seven nodes?



$$\begin{aligned} \text{(b)} \quad & 17 + 20 + 27 + 25 + 26 + 24 \\ & = 139 \text{ km} \end{aligned}$$

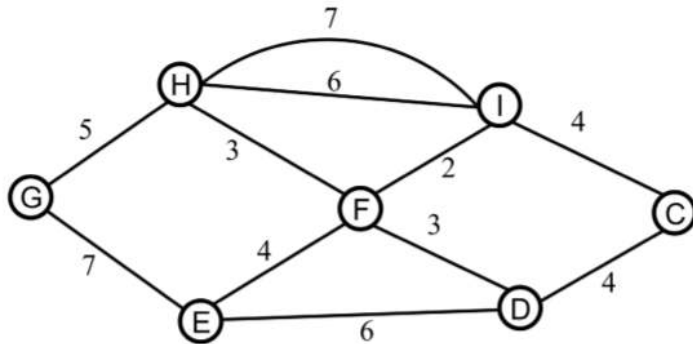


Q 66

James Busby

The diagram below shows the seven suburbs of a city and the network of public transport links between them.

The amounts represent the cost (in dollars) of travel using that link.



Calculate the minimum cost of travelling from G to C and list the path which achieves this cost.

$$5 + 3 + 2 + 4 = 14$$

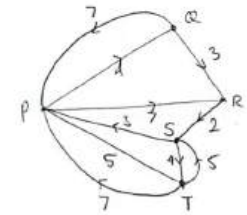
G-H-F-I-C

Q 67

James Busby

A directed network is represented by the table below.

		To				
		P	Q	R	S	T
From	P	-	4	7	-	5
	Q	7	-	3	-	-
	R	-	-	-	2	6
	S	3	-	-	-	4
	T	7	-	-	5	-

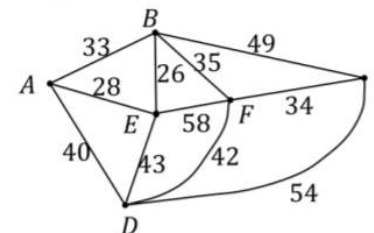


Draw the network in the space below.

Q 68

Cambridge

Prim's algorithm, beginning with vertex B, will be used to find the minimal spanning tree for the network below.



Which edge will be added last?

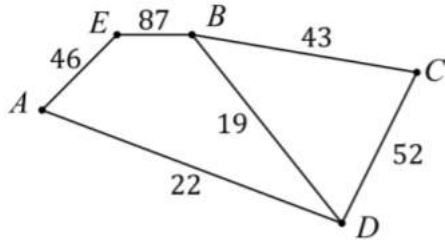
- (A) CF (C) ED
(B) AB (D) AD

D



Q 69

Cambridge



What is the length of the shortest path from E to C ?

- (A) 120
- (B) 130
- (C) 158
- (D) 186

Q 70

Cambridge

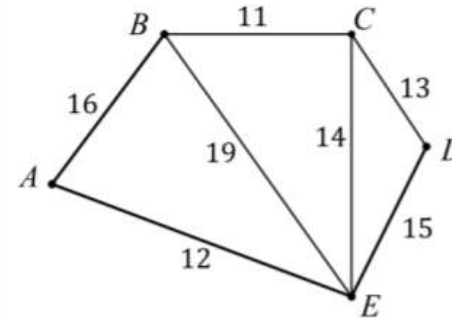
There are five towns labelled A, B, C, D and E . The table below shows which towns are linked by the roads and the length of each one in kilometres.

	A	B	C	D	E
A	–	16	–	–	12
B	16	–	11	–	19
C	–	11	–	13	14
D	–	–	13	–	15
E	12	19	14	15	–

- (a) Represent the table shown above as a weighted network.
- (b) What is the degree of vertex E ?
- (c) What is the shortest path from town A to town C ?

11(a)

Weight edge: $AB = 16, AE = 12, BC = 11, BE = 19, CD = 13, CE = 14, DE = 15$



11(b)

There are four edges connected to vertex E .
 $\therefore$ Degree of vertex E is 4.

11(c)

Shortest path from A to C .
 $A-E-C$

Q 71

Cambridge

Explain the difference between finding a minimal spanning tree and a shortest path?

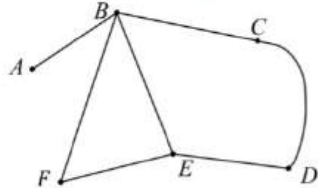
A minimal spanning tree connects all the vertices together with the minimum total weighting for the edges. In contrast the shortest path is the path where the sum of the weights of its edges is minimised. It need not necessarily contain all the other vertices.



Q 72

Cambridge

Consider the following network.

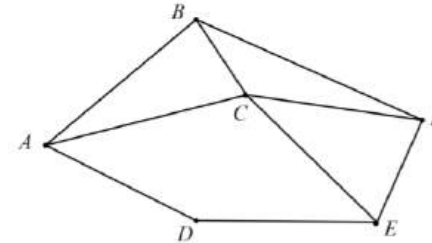


- How many edges are there?
- How vertices have degree 2?
- Is this a connected graph? Justify your answer.
- List three different cycles that begin at vertex B ?

26(a)	There are 7 edges: $AB, BF, BE, BC, CD, EF, ED$
26(b)	Three vertices with degree 2 (vertex C, D and F)
26(c)	Yes, it is a connected graph. Connected graph if every vertex in the graph is accessible from every other vertex in the graph along a path formed by the edges of the graph.
26(d)	Cycle is a walk with no repeated vertices or edges that starts and ends at the same vertex. Cycles: $BEFB, BCDEB, BCDEFB$

Q 73

Pymble



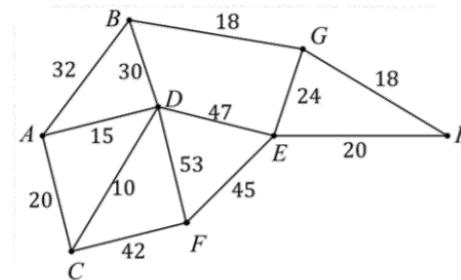
Which of the following represents a cycle in the above network diagram?

- $D-A-C-F-D$
- $C-F-E-D-C$
- $B-A-D-E-C-B$
- $E-F-C-A-B-E$

Q 74

Pymble

Lucy owns a holiday park and has been quoted \$45 per metre to provide connecting paths to each of the cabins, as shown below. All measurements are in metres.



What is the least it will cost Lucy to provide each cabin with at least one path that connects all the cabins?

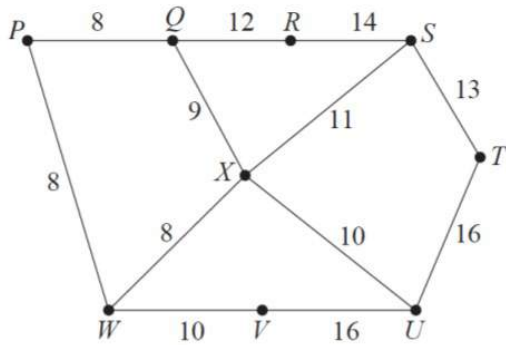
- \$6435
- \$6885
- \$7470
- \$8820



Q 75

Pymble

The network diagram shows the time (in minutes) taken to travel between towns.



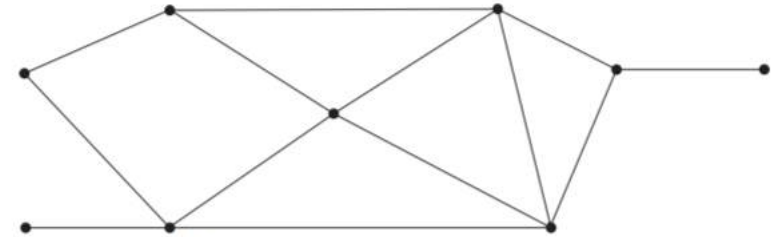
Find the shortest time it would take to travel from P to T .

Shortest time = $8 + 8 + 11 + 13$
= 40 minutes

Q 76

NEAP

A network diagram is shown.



The number of vertices of odd degree is

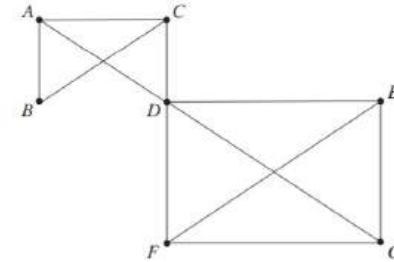
- A. 2 C. 6
B. 4 D. 9

B

Q 77

NEAP

A network diagram is shown.



C

Which of the following could be both a spanning tree AND a path for the network diagram?

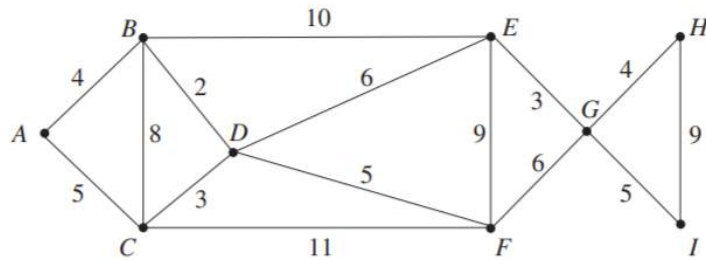
- A. B. C. D.



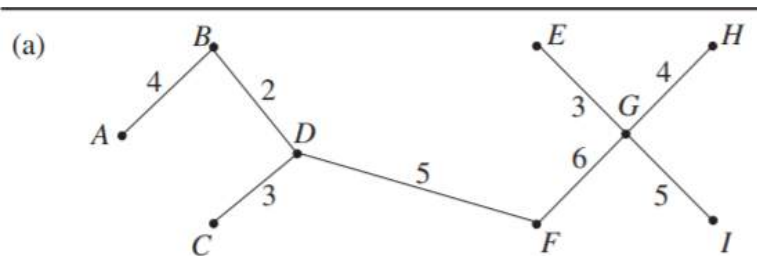
Q 78

NEAP

The network diagram shows the cost, in hundreds of dollars, to connect electrical cables to multiple points in a building.



- (a) Draw the minimum spanning tree for this network in the space below.
- (b) Using the minimum spanning tree, calculate the minimum cost to connect a cable to every point in the building.



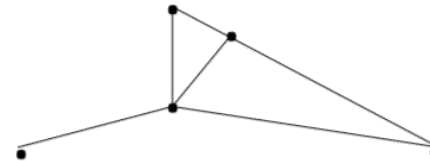
(b) $4 + 2 + 3 + 5 + 6 + 3 + 4 + 5 = 32$

The network diagram shows the cost in hundreds of dollars. Therefore, the minimum cost is \$3200.

Q 79

NEAP

The diagram shows a network



What is the sum of all the degrees of the vertices in this network?

- (A) 5 (C) 11
(B) 9 (D) 12

C

Q 80

Trials

The costs (in \$) of connecting various locations on a school campus with computer cable are given in the table below. A blank space indicates no direct connection.

	A	B	C	D	E
A	—	4000		5000	3000
B	—	—	1500	2200	4500
C	—	—	—	2200	1500
D	—	—	—	—	2500

Draw a network to represent this situation, showing the cost of connection along each arc.

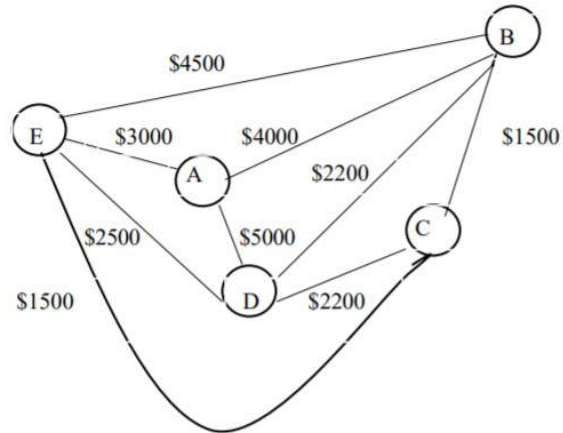


networks selective



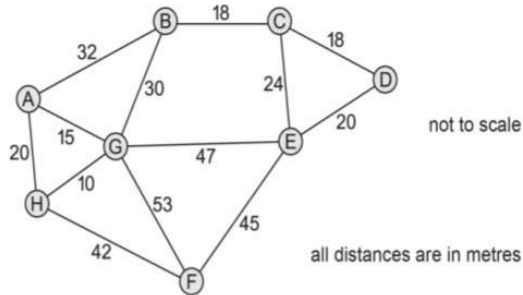
NAME: _____

Standard



Q 81

Trials



The above network shows connecting paths for a cable network. It costs \$65 per metre to lay the cable. Use a minimum spanning tree to determine the least cost to connect this network.

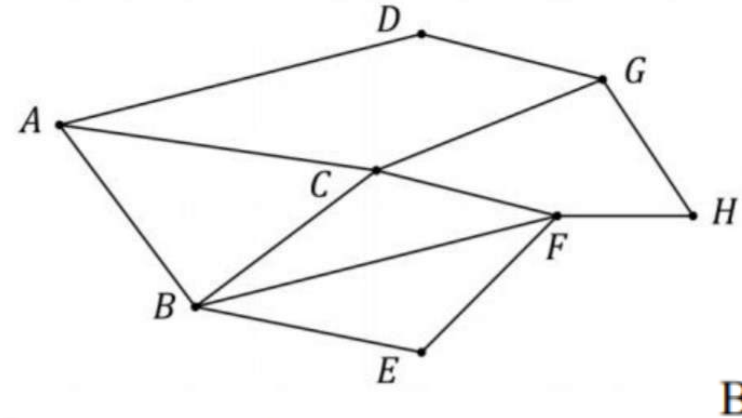
Minimum spanning tree : $15 + 10 + 42 + 30 + 18 + 18 + 20 = 153 \text{ m}$

Cost : $153 \text{ m} \times \$65 = \9945

Q 82

QATs

What is the degree of the vertex F ?



A.

3

C.

5

B.

4

D.

8

Q 83

QATs

In a town, there are four petrol stations A , B , C and D . The table shows the length, in metres, of roads connecting these petrol stations.

	A	B	C	D
A	-	550	-	180
B	550	-	400	250
C	-	400	-	380
D	180	250	380	-

A petrol tanker needs to visit each station.

What is the shortest distance it can travel if it starts at Station A ?

A.

830 m

C.

1 180 m

B.

1 110 m

D.

1 380 m

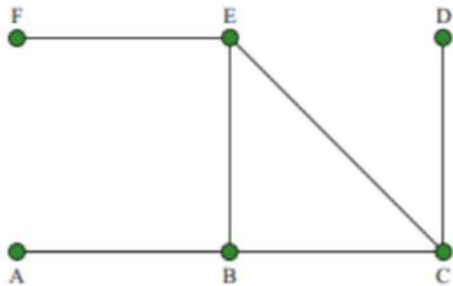


Q

ABFDECA

$$30 + 70 + 90 + 80 + 120 + 40 = 430 \text{ m}$$

C

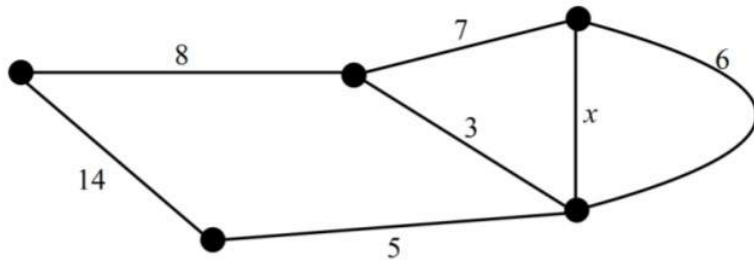


- A. AB
- B. EF
- C. BC
- D. CD

Q 85

QATs

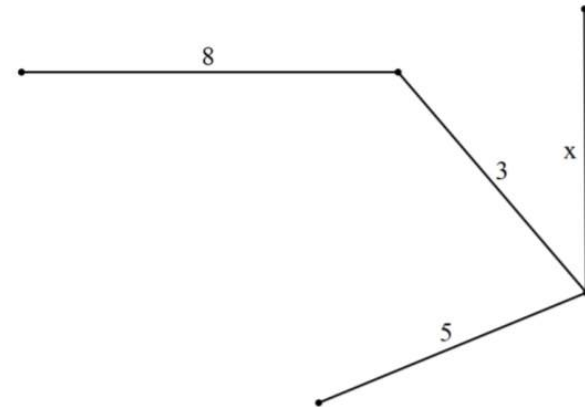
The minimum spanning tree for the below network **includes the edge x** .



- (a) Draw the minimum spanning tree for the above network.
- (b) If x is included in the minimum spanning tree, what can you say about the value of x ?

Question 20

20 (a)



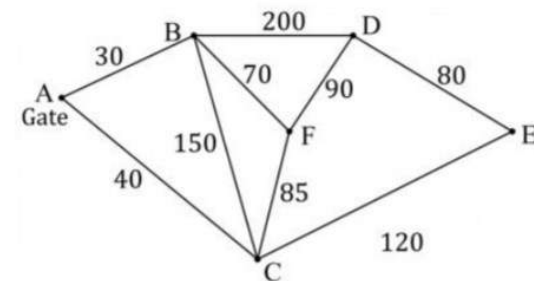
20 (b)

x must be less than 6

Q 86

QATs

The network diagram below shows an orienteering challenge with five tasks (B, C, D, E, F) that are joined by paths. You must start and finish the challenge at the gate (A). The numbers show the distance along the paths in metres.



By first stating the path, calculate the minimum distance a team would have to run to complete every task in the challenge, starting and ending at the gate.

ABFDECA

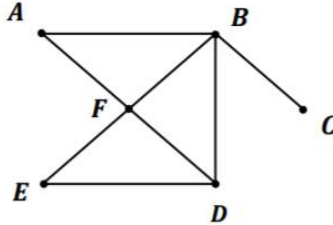
$$30 + 70 + 90 + 80 + 120 + 40 = 430 \text{ m}$$



Q 87

Hunter Hills

What is the degree of vertex B in this network diagram?

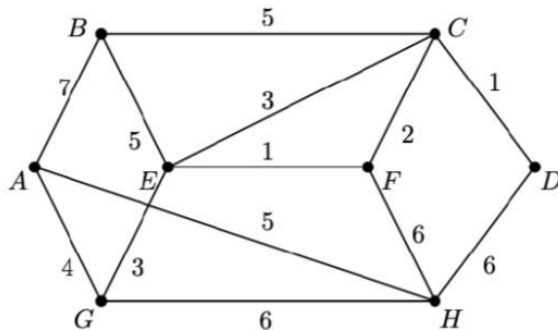


- A 4
- B 3
- C 2
- D 1

Q 88

Hunter Hills

This network diagram shows weighted edges (in metres) connecting various vertices.



NOT TO
SCALE

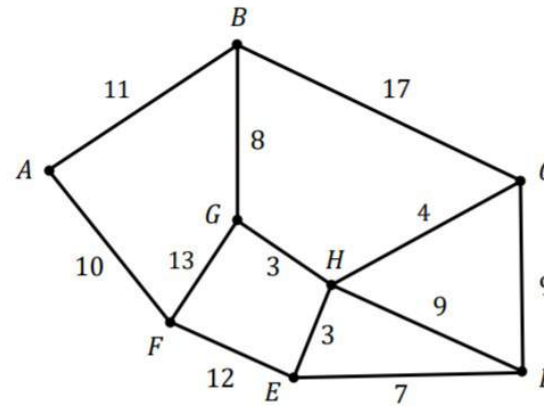
What is the shortest distance from A to D ?

- A 11 m
- B 13 m
- C 15 m
- D 16 m

Q 89

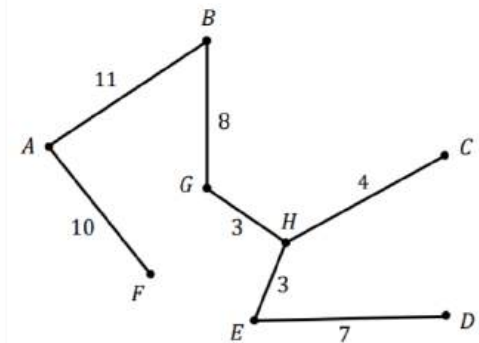
Hunter Hills

The network diagram shows the cost, in thousands of dollars, to connect internet cables between various access points.



- (a) Draw the minimum spanning tree in the space provided.
- (b) Use the minimum spanning tree to determine the minimum amount that must be spent to connect all access points.

23 a)



There is only one MST which is any isomorphic to the one above.

23 b)

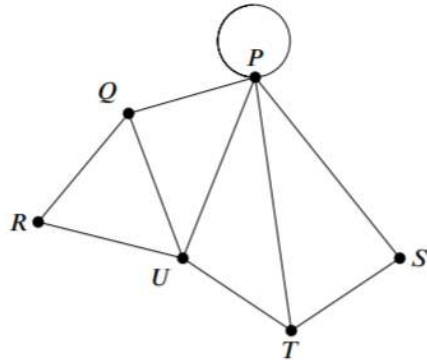
Weight = 46
 $\therefore$ Cost = $46 \times \$1000 = \46000



Q 90

Kambala

The diagram below shows a network with a loop at vertex P .



D

What is the degree of vertex P in the network below?

- A. 3 B. 4 C. 5 D. 6

Q 91

Kambala

A network has x edges. Which expression below gives the sum of the degrees of the vertices in the network?

- A. x B. $x + 1$ C. $2x$ D. $2x + 1$

Q 92

Kambala

A network contains 50 vertices. How many edges are required to form a spanning tree?

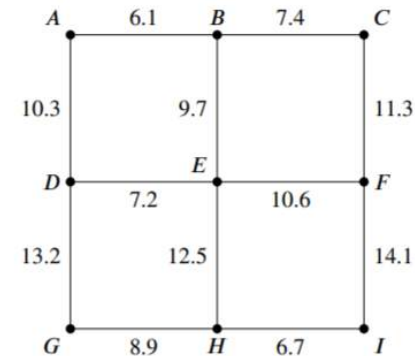
- A. 24 B. 25 C. 49 D. 50

C

Q 93

Kambala

The following network shows the length, in kilometres, of roads connecting nine villages. A programme of resurfacing some roads is undertaken to ensure that each village can access all other villages along a resurfaced road.



D

Using a minimum spanning tree, what length of road, in kilometres, will connect all nine villages while keeping the amount of road to be resurfaced to a minimum?

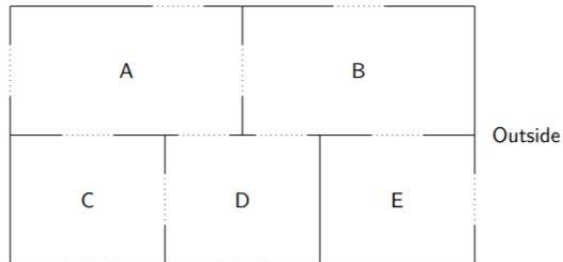
- A. 66.9 B. 67.6 C. 67.9 D. 69.1



Q 94

Kambala

The diagram below shows a floor plan of a new building with five rooms. Each room has a number of doors to another room or to the outside.



- (a) Complete the matrix below, indicating the number of possible doors between each room and an adjacent room or the outside (O).

	A	B	C	D	E	O
A	—					
B		—				
C			—			
D				—		
E					—	
O						—

- (b) Bob the builder needs to inspect each door. He wants to start outside the building, then check every door, and finish outside the building again, passing through each doorway once and once only. Is this possible? Use your answer to (a) or the floor plan to justify your answer.

(a)

	A	B	C	D	E	O
A	—	1	1	1	0	2
B	1	—	0	1	1	1
C	1	0	—	1	0	1
D	1	1	1	—	0	1
E	0	1	0	0	—	1
O	2	1	1	1	1	—

- (b) Degree of room A = 5
Degree of room B = 4
Degree of room C = 3
Degree of room D = 4
Degree of room E = 2
Degree of outside = 6

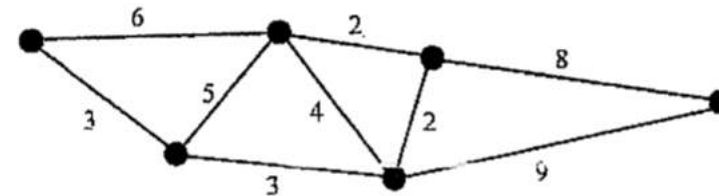
It is not possible to Bob to start outside, then inspect every doorway, and finish outside the building. For this to occur, the degree for each of the rooms (vertices) would need to be even. Here we have two odd vertices.

Q 95

CSSA

A weighted network is given below.

What is the total length of the minimum spanning tree?



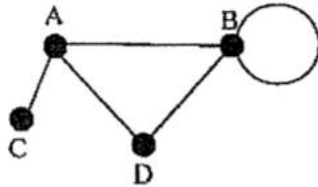
- (A) 15
(B) 18
(C) 21
(D) 22



Q 96

CSSA

A simple network is shown below.



Complete the table of vertex degrees for the network.

Vertex	A	B	C	D
Degree				

Vertex	A	B	C	D
Degree	3	4	1	2